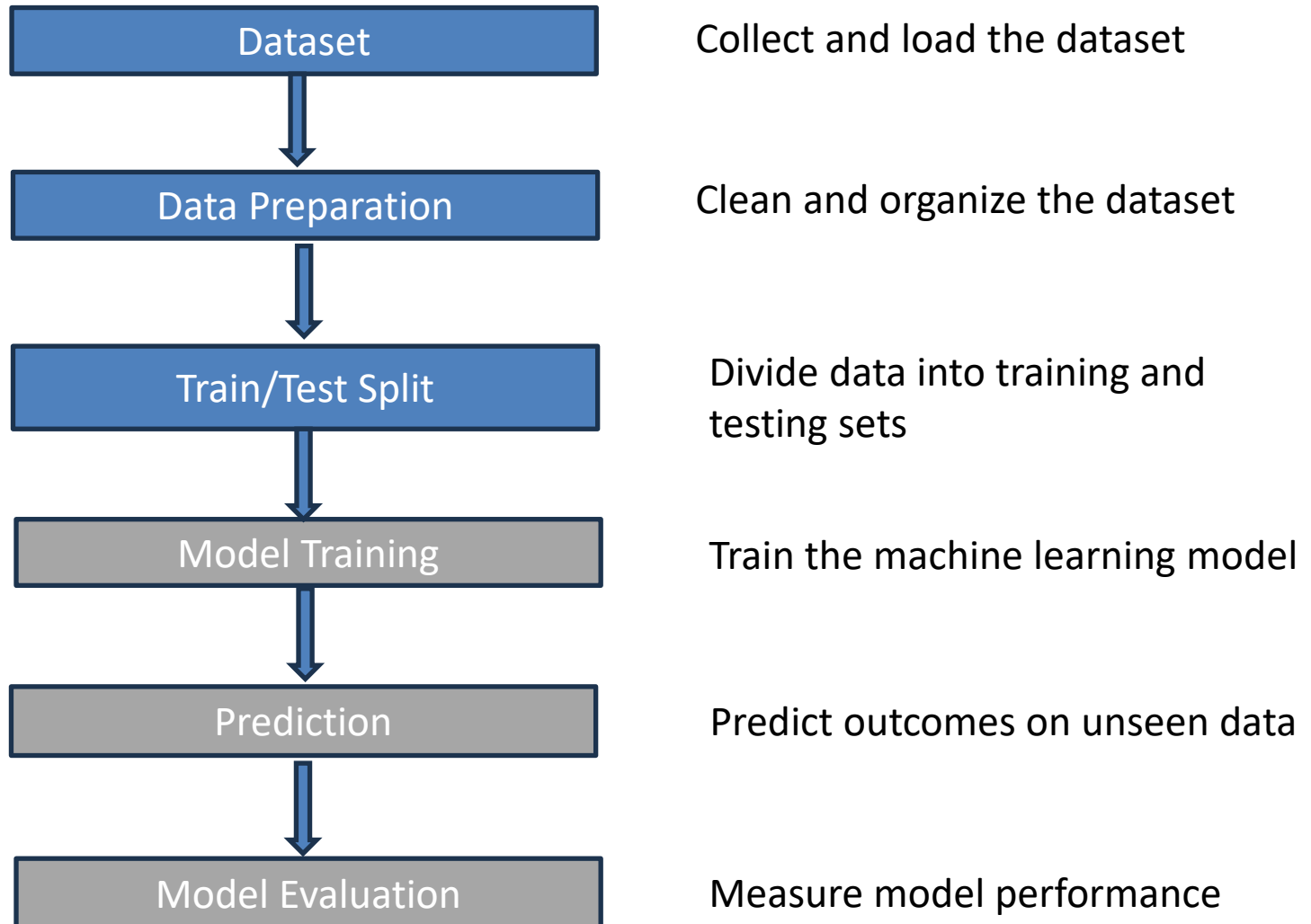


Topics Covered

- Machine Learning Workflow
- Understanding a Dataset
- Mathematical representation of a dataset
- Training, Validation and Testing Datasets
- Reading a Dataset using Pandas
- Code Implementation

Machine Learning Workflow



Understanding a Dataset

- ❑ A dataset is a collection of data samples used to train, validate, and test a machine learning model.
- Each row represents one data sample (observation/instance).
- Each column represents one feature (attribute/variable).
- The last column is usually the target (class label).
- Machine learning models learn the relationship between features (X) and the target variable (y).
- Dataset components
 - At1 = Feature 1
 - At2 = Feature 2
 - Class = Target Label

At1	At2	Class
1.14	-0.114	b'-1.0'
-1.52	-1.15	b'1.0'
-1.05	0.72	b'-1.0'
-0.916	0.397	b'1.0'
-1.09	0.437	b'1.0'
-0.584	0.0937	b'1.0'
1.83	0.452	b'-1.0'

Mathematical Representation of a Dataset

Feature Matrix (X)

$$X \in \mathbb{R}^{n \times d}$$

- X = Collection of all input features
- $\mathbb{R}$ = set of real numbers
- n = Number of data samples (rows)
- d = Number of features (columns)

Target Vector (y)

- $y \in \{-1, 1\}$
- y = Target labels
- Banana dataset: $\{-1, 1\}$

After Train-Test Split

- $(X, y) \rightarrow (X_{train}, X_{test}, y_{train}, y_{test})$
- **X_train** → Features used for training.
- **X_test** → Features used for testing.
- **y_train** → Training labels.
- **y_test** → Testing labels.

Training, Validation and Testing Datasets

❑ Training Dataset

- Used to train the machine learning model.
- Usually contains the largest portion of the dataset.

❑ Validation Dataset

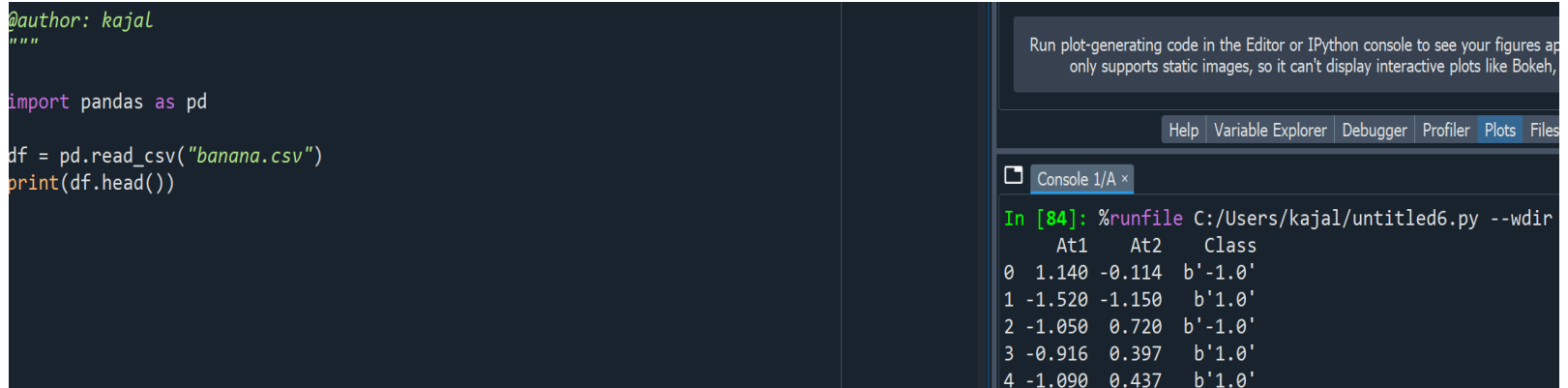
- Used to tune hyperparameters and improve the model.
- Used to improve the model before final testing.

❑ Testing Dataset

- Used to evaluate the final model performance.
- Contains unseen data that was not used during training.

Reading a Dataset using Pandas

- ❑ Pandas is a Python library used to read, organize, and analyze datasets.



The screenshot shows a Jupyter Notebook interface. The left pane contains the following Python code:

```
@author: kajal
"""

import pandas as pd

df = pd.read_csv("banana.csv")
print(df.head())
```

The right pane shows the output of the code. At the top, there is a message: "Run plot-generating code in the Editor or IPython console to see your figures as only supports static images, so it can't display interactive plots like Bokeh,". Below this, there are tabs for "Help", "Variable Explorer", "Debugger", "Profiler", "Plots", and "Files". The "Console" tab is active, showing the output of the code:

```
In [84]: %runfile C:/Users/kajal/untitled6.py --wdir
         At1    At2    Class
0    1.140 -0.114  b'-1.0'
1   -1.520 -1.150  b'1.0'
2   -1.050  0.720  b'-1.0'
3   -0.916  0.397  b'1.0'
4   -1.090  0.437  b'1.0'
```

Pandas converts the dataset into a DataFrame, making it easier to explore and process the data.

Code Implementation

```
14 import pandas as pd
15 import numpy as np
16 from sklearn.model_selection import train_test_split
17
18 filename = "glass.csv"
19 df = pd.read_csv(filename)
20 print("=" * 60)
21 print("DATASET INFORMATION")
22 print("=" * 60)
23 print("Dataset Used :", filename)
24
25 print("\nFirst 5 Rows of Dataset")
26 print("-" * 60)
27 print(df.head())
28
29 rows = df.shape[0]
30 columns = df.shape[1]
31
32 print("\nNumber of Rows      :", rows)
33 print("Number of Columns   :", columns)
34
35 label = df.columns[-1]
36
37 classes = df[label].unique()
38
39 print("Number of Classes :", classes)
40 print("\nDataset Shape :", df.shape)
41
42 X = df.drop(label, axis=1)
43 y = df[label]
44
45 print("\nFeature Columns")
46 print(X.columns.tolist())
47
48 print("\nTarget Column:")
49 print(label)
```

Run plot-generating code in the Editor or IPython console to see your figures appear here. This pane

Help Variable Explorer Debugger Profiler Plots Files

Console 1/A x

```
4 1.51742 13.27 3.62 1.24 73.08 0.55 8.07 0.0 0.0 1
```

Number of Rows : 214
Number of Columns : 10
Number of Classes : 6

Dataset Shape : (214, 10)

Feature Columns
['RI', 'Na', 'Mg', 'Al', 'Si', 'K', 'Ca', 'Ba', 'Fe']

Target Column:
Type

=====

MATHEMATICAL REPRESENTATION

=====

X belongs to $R^{(214 \times 9)}$
y belongs to : [1 2 3 5 6 7]

Number of Samples (n) : 214
Number of Features (d): 9

=====

TRAIN-TEST SPLIT (70 : 30)

=====

X_train Shape : (149, 9)
X_test Shape : (65, 9)
y_train Shape : (149,)
y_test Shape : (65,)

IPython Console History